

1. "Integrated navigation, guidance and control system and validation" (Barua et al., 2018) - RLV-TD Mission

How Navigation was Achieved: Navigation relies on a GPS-aided inertial navigation system (GAINS), where a Mission Management Computer processes sensor data to generate closed-loop guidance and control. A Kalman filter estimates position and velocity errors, which are then used to correct the navigation data in a feed-forward manner. **Sensors Used and Why:**

- **Inertial Navigation System (INS):** Uses three gyros in an ortho-skewed configuration and five accelerometers to provide fail-operational redundancy for calculating attitude, position, and velocity.
- **Global Positioning System (GPS):** A stand-alone INS cannot meet the strict accuracy needed for landing (50 m position, 0.5 m/s velocity), so it is aided by the GPS.
- **Radar Altimeter (RA):** Enhances altitude accuracy to 1 m.
- **Flush Air Data System (FADS):** Measures air data parameters like angle of attack, sideslip, and Mach number.
- **Ceramic Servo Accelerometer Package (CSAP):** A triad-hexad geometry of six accelerometers used to generate lateral acceleration data specifically for the descent control.

Vision-Based Sensors: None. The system relies entirely on inertial, satellite, radar, and aerodynamic sensors.

Altitude Dependence and Sensor Favoring: Sensor usage changes significantly with altitude. The **Radar Altimeter is specifically favored below an altitude of 2 km** during the landing phase to correct navigation altitude. The **CSAP is heavily favored during the descent phase**, particularly when wind effects are prominent and estimations of aerodynamic angles become less accurate. During "Phase 3" of descent (when altitude drops **below 20 km** and velocity below Mach 2), wind velocities heavily influence aerodynamic angle estimations; therefore, the vehicle stops using aerodynamic angles for control and instead commands accelerations sensed directly by the CSAP.

2. "Preliminary Design of the Hybrid Navigation System (HNS) for the CALLISTO RLV Demonstrator" (Schwarz et al., 2019)

How Navigation was Achieved: CALLISTO utilizes a Hybrid Navigation System (HNS) that overcomes the continuous integration drift of purely inertial systems by fusing high-frequency inertial measurements with low-frequency absolute measurements from external sensors.

Sensors Used and Why:

- **Tetra-axial IMU:** Includes four gyroscopes and four accelerometers to maintain continuous high-rate measurements and tolerate the failure of a single sensor.
- **Differential GNSS (DGNSS):** Standard GNSS cannot meet the 1.0 m horizontal landing accuracy requirement. The DGNSS uses multi-constellation tracking (GPS/Galileo) and a ground-based reference station to instantly correct satellite errors via telemetry.
- **Radar Altimeters:** Two altimeters are placed on the aft bay to meet the strict 0.5 m vertical accuracy requirement for landing.

Vision-Based Sensors: Evaluated but rejected. Visual navigation systems (like AprilTags or QR codes) were explicitly considered but discarded due to high operational efforts for camera calibration, sensitivity to changing lighting, and because the **engine exhaust plume would significantly interfere with the camera.**

Altitude Dependence and Sensor Favoring: The **Radar Altimeters** are strictly utilized **below 500 m altitude**. Analyses proved that without the radar updates, the vertical position uncertainty would exceed mission requirements at 100 m altitude, making the radar altimeter critically favored over purely satellite-based systems during the final descent.

3. "Overview of Flight Guidance, Navigation, and Control for the DLR Reusability Flight Experiment (ReFEx)" (Schwarz et al., 2019)

How Navigation was Achieved: Navigation relies on a centralized "HNS Box" that acts as a redundant, fail-operational data fusion center, combining an over-determined inertial setup with GPS, altimeters, and optical sensors. **Sensors Used and Why:**

- **IMU:** 4 COTS accelerometers and 4 modified gyroscopes. Because the vehicle spins at 4 Hz during ascent, the gyros were heavily modified to measure up to $\pm 1,500$ °/s.
- **GPS (Phoenix-HD):** Uses a heat-shielded, **wrap-around antenna** to ensure spin-insensitivity and constant signal reception while the vehicle rolls.
- **Laser & Radar Altimeters:** Used to precisely calculate altitude over ground to trigger a final flare maneuver prior to touchdown.

Vision-Based Sensors: Yes, Sun Sensors are used. ReFEx is equipped with four digital two-axis COTS Sun sensors spaced around the vehicle. The high spin rate during ascent builds up significant attitude estimation errors; the Sun sensors (each with a 120-degree field of view) are explicitly used to track the sun and correct these attitude errors after the vehicle despin maneuver.

Altitude Dependence and Sensor Favoring:

- **Altimeter Handoff:** The **Laser Altimeter** operates from **1,500 m down to 100 m** altitude. At exactly **100 m over ground**, the **Radar Altimeter starts its operation** (with an 800 Hz update rate) to guide the final flare maneuver.

- **Sun Sensor Limitations:** The Sun sensors can only operate effectively at **altitudes higher than 40 km**, where the "atmospheric glow" of the sky is low enough. Consequently, as the vehicle drops to lower altitudes during descent, the Sun sensors stop operating due to the atmosphere, and the vehicle gradually loses this attitude improvement, falling back to IMU accuracy.
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4. "Hybrid Navigation System for Spaceplanes, Launch and Re-Entry Vehicles" (Theil et al., 2009) - SHEFEX-2 Mission

How Navigation was Achieved: Navigation is performed via a loosely coupled Kalman filter that merges a fast (500 Hz) strapdown inertial routine with lower-rate GPS (1 Hz) and star tracker (2 Hz) updates. **Sensors Used and Why:**

- **IMU:** Samples at 500 Hz to minimize sculling errors caused by the vehicle spinning up to 2.5 Hz.
- **Phoenix-HD GPS:** Aids signal acquisition and trajectory monitoring.
- **Star Tracker (STR):** Launch vibrations cause the IMU to drift by 2 to 3 degrees, but the flight control experiment strictly requires 0.17 degrees of accuracy. The STR is used to fix this massive drift.

Vision-Based Sensors: Yes, a Star Tracker camera is used. DLR custom-built an off-the-shelf camera and lens system. The onboard software uses dark image subtraction to ignore bad pixels, a centroiding algorithm to find stars, and a geometric algorithm comparing inter-star angles to a database to autonomously calculate the vehicle's orientation.

Altitude Dependence and Sensor Favoring: The **Star Tracker is only operational during the high-altitude coast phase** (after the vehicle has spun down and just before atmospheric entry). During the highly dynamic propelled flight phases, only the INS is used because launch vibrations are too intense for other sensors. Furthermore, **GPS blackouts are explicitly expected during the atmospheric entry phase.**

5. "Architectures of Hybrid Navigation Systems (HNS) for Reusable Space Transportation Systems" (Schwarz et al., 2019)

(Note: This paper acts as an architectural comparison between the CALLISTO and ReFEX missions.)

How Navigation was Achieved: Both vehicles rely on a distributed GNC architecture featuring a centralized, double-modular hot-redundant HNS Box. This box continuously integrates inertial data and applies corrections from external sensors. **Sensors Used and Why:** The paper highlights how the sensor suite changes based on the vehicle:

- CALLISTO (Vertical landing) requires extreme landing accuracy and relies on **DGNSS and Radar Altimeters**.
- ReFEx (Horizontal landing) utilizes a single-constellation **GNSS with a wrap-around antenna, Laser/Radar altimeters, and Sun sensors**.

Vision-Based Sensors: Re-iterates that ReFEx incorporates Sun sensors to correct attitude drift, while CALLISTO does not use inflight optical sensors.

Altitude Dependence and Sensor Favoring: This paper confirms the altitude handoffs detailed earlier: CALLISTO relies on Radar Altimeters specifically **below 500 m**, while ReFEx uses the **Laser Altimeter down to 100 m**, at which point it hands over to the **Radar Altimeter**. It also confirms that ReFEx Sun sensors are favored at high altitudes but lose their effectiveness at lower altitudes due to atmospheric interference.

6. "Towards a Reusable First Stage Demonstrator: CALLISTO - Technical Progresses & Challenges" (Krummen et al., 2021)

How Navigation was Achieved: Navigation focuses on refining sensor fusion for the vertical landing phase using an advanced multi-constellation, multi-frequency GNSS subsystem paired with absolute altitude measuring tools. **Sensors Used and Why:**

- **DGNSS & RTK:** The paper notes a shift toward evaluating a Real-Time Kinematic (RTK) based approach using the GNSS ground reference station to determine the exact flight altitude as an alternative to radar.
- **Radar Altimeter:** A prototype was successfully tested by firing radar signals directly through the hot engine exhaust plume during static firing tests. It proved that engine plumes do not have a significant negative impact on the radar's ability to find the landing pad.

Vision-Based Sensors: No in-flight vision sensors are used. However, the paper reveals that an **external alignment concept based on optical measurements** was developed for use **on the launch pad before launch** to improve the initial attitude estimation accuracy.

Altitude Dependence and Sensor Favoring: The paper specifically isolates the **"final flight phase"** as the regime where either the RTK-based approach or the Radar Altimeter must be favored to accurately determine the flight altitude above the landing pad.

7. "End-to-End GNC Solution for Reusable Launch Vehicles" (Guadagnini et al., 2025) - RETALT1

How Navigation was Achieved: Navigation relies on sensor fusion combining IMUs and (D)GNSS receivers to estimate the vehicle's state. The GNC system utilizes a highly advanced onboard Sequential Convex Programming (SCP) algorithm to compute fuel-optimal trajectories in real-time. **Sensors Used and Why:**

- **IMU & (D)GNSS:** Form the core state estimator across all flight phases (ballistic arc, re-entry burn, aerodynamic descent, and landing burn).
- **Altimeter & FADS:** Flush Air Data Sensing and altimeters are integrated to further improve the accuracy of the state estimation as the vehicle physically approaches the landing site.

Vision-Based Sensors: None. The architecture presented completely relies on IMU, GNSS, radar/laser altimetry, and air data sensors.

Altitude Dependence and Sensor Favoring:

- **High-Altitude Ballistic Phase:** At high altitudes after separation, neither thrust nor aerodynamic forces are available; the vehicle relies strictly on Reaction Control Systems (RCS) to perform a flip-over maneuver.
- **Re-entry Burn Phase:** This phase occurs at a relatively high altitude (>65 km). Because the atmosphere is so thin and the engines are switched on (which reduces the drag coefficient), **aerodynamic forces are entirely neglected** in the guidance dynamics to speed up computational solving.
- **Landing Burn Phase:** As the vehicle drops to low altitudes, **altimeters and (F)ADS are explicitly favored/introduced** to improve accuracy when approaching the landing barge. Additionally, because of the low altitude and low velocity in this phase, wind speed profiles are explicitly integrated into the optimization problem.

Vision-based sensors are generally considered unreliable for Reusable Launch Vehicle (RLV) navigation primarily due to the harsh environmental conditions of rocket flight, particularly the interference from the engine's exhaust plume and varying lighting conditions. Classical sensors—such as Radar Altimeters, Inertial Measurement Units (IMUs), and Global Navigation Satellite Systems (GNSS)—are preferred because they bypass these limitations to offer continuous, robust measurements.

Here is a detailed explanation of why vision-based sensors are inadequate and why classical sensors are favored:

Drawbacks of Vision-Based Sensors

- **Engine Plume and Exhaust Interference:** During the critical landing phase, RLVs rely on retro-propulsion. Visual navigation systems (like AprilTag or QR code readers) require a clear line of sight to the landing pad, which is significantly obstructed by the dense, hot exhaust plume generated by the rocket engine.
- **Sensitivity to Environmental Conditions:** Optical systems are heavily dependent on ideal environmental states. They struggle with changing lighting conditions, dust, and vapor. Even specialized optical sensors like Star Trackers or Sun Sensors face operational limits due to "atmospheric glow" and can be easily blinded if bright sources (like the Earth or Sun) enter their field of view, requiring strict exclusion angles and protective baffles.
- **Vulnerability to High Dynamics (Image Smear):** RLVs often undergo highly dynamic maneuvers, such as spinning or rolling at high rates. These rapid movements cause "image smear" in cameras, which restricts the time windows during which the sensors can provide reliable data, or forces the use of very short exposure times that reduce the signal-to-noise ratio. For example, Sun sensors on the ReFEx mission lose their utility entirely during high spin rates.
- **High Operational Effort:** Implementing a functional visual navigation system demands significant operational effort, including precise and constant camera calibration, as well as the calibration of additional Ground Support Equipment (GSE).

Why Classical Sensors are Better

Instead of optical cameras, modern RLV navigation relies on a **Hybrid Navigation System (HNS)** that fuses data from classical sensors, combining their strengths to achieve pinpoint accuracy.

- **Radar Altimeters are Plume-Resistant:** Unlike cameras and laser altimeters, radar altimeters use electromagnetic radio waves that can effectively penetrate dust, vapor, and hot engine exhaust plumes without significant degradation. This makes them the most feasible and robust solution for measuring the exact altitude over the landing pad during the final touchdown phase.
- **IMUs are Immune to External Interference:** An Inertial Measurement Unit (IMU) acts as the core of the navigation system by measuring angular velocities and accelerations

internally. Because IMUs do not rely on any external references or line-of-sight, they are completely immune to environmental obscuration, lighting changes, signal jamming, or spoofing.

- **GNSS/DGNSS Provide Absolute Robustness:** To correct the natural drift of the IMU over time, vehicles use GNSS (and Differential GNSS). DGNSS provides an absolute horizontal positioning accuracy of better than 1 meter by utilizing ground-based reference stations to instantly correct signal errors. This provides highly accurate, weather-independent absolute positioning that does not require the vehicle to "see" the landing zone.

In summary, while vision sensors are flexible in benign environments, the extreme dynamics, dust, and engine exhaust associated with rocket launches and landings make them too fragile. Classical sensors offer the **all-weather, plume-penetrating, and environmentally independent reliability** required to safely land a rocket.